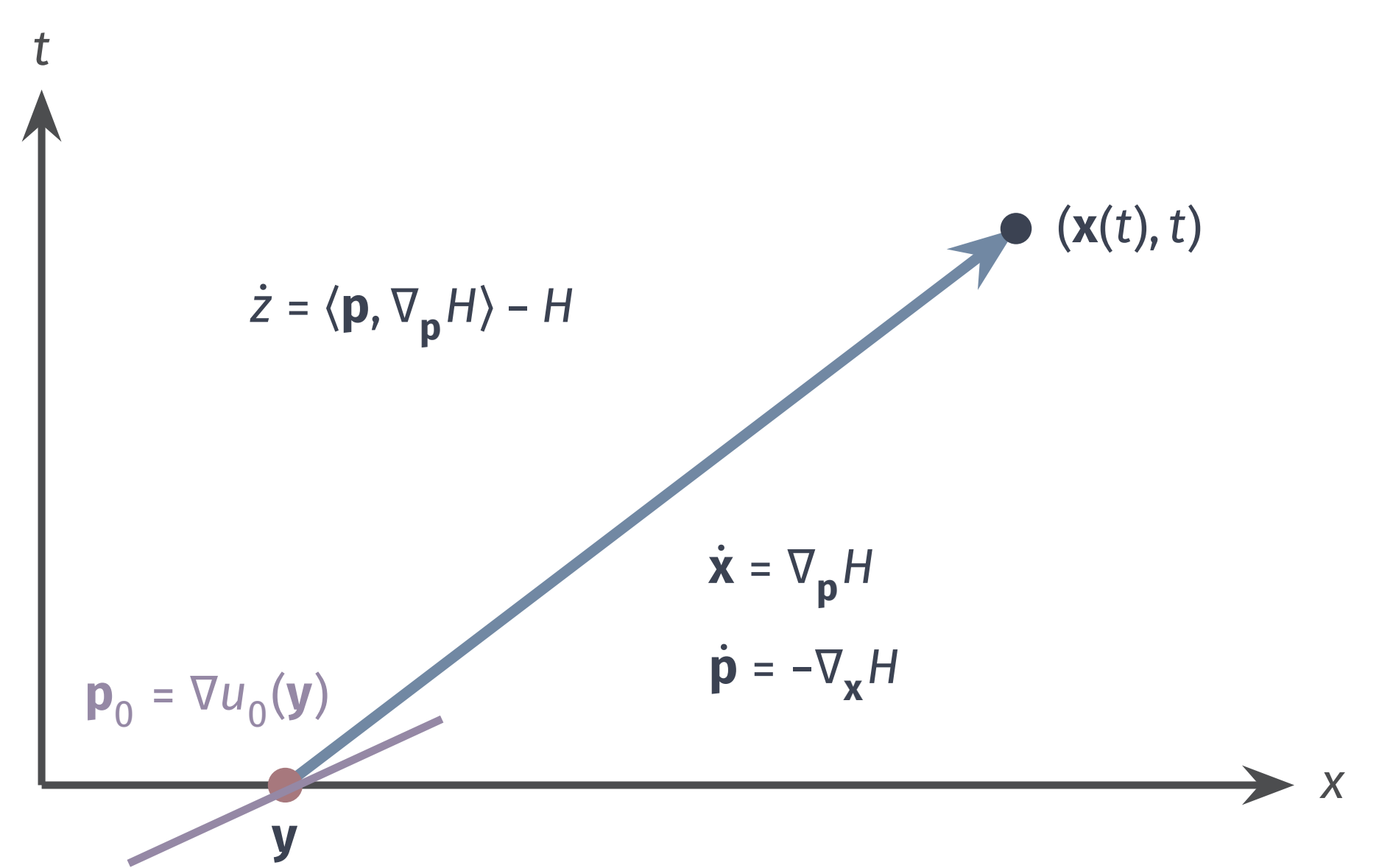
t
$(\mathbf{x}(t), t)$
$\dot{z} = \langle \mathbf{p}, \nabla_{\mathbf{p}} H \rangle - H$
$\dot{\mathbf{x}} = \nabla_{\mathbf{p}} H$
$\dot{\mathbf{p}} = -\nabla_{\mathbf{x}} H$
$\mathbf{p}_0 = \nabla u_0(\mathbf{y})$
x
y